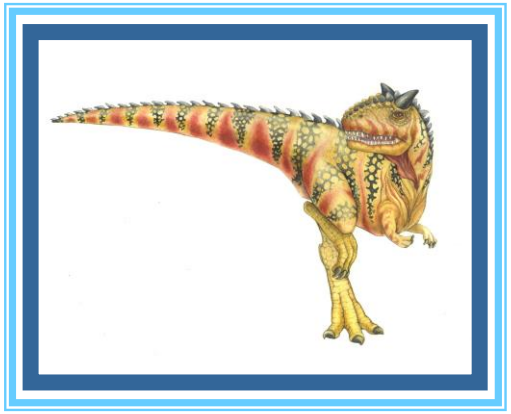
Chapter 15: File System Internals



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Chapter 15: File System Internals

File Systems

File-System Mounting

Partitions and Mounting

File Sharing

Virtual File Systems

Remote File Systems

Consistency Semantics

NFS

Objectives

a

File System

a

b

Partitions and Mounting

Partition can be a volume containing a file system (“cooked”) or **raw** – just a sequence of blocks with no file system

Boot block can point to boot volume or boot loader set of blocks that contain enough code to know how to load the kernel from the file system

- Or a boot management program for multi-os booting

Root partition contains the OS, other partitions can hold other Oses, other file systems, or be raw

- Mounted at boot time
- Other partitions can mount automatically or manually

At mount time, file system consistency checked

- Is all metadata correct?
  - If not, fix it, try again
  - If yes, add to mount table, allow access

File Sharing

a

b

Virtual File Systems

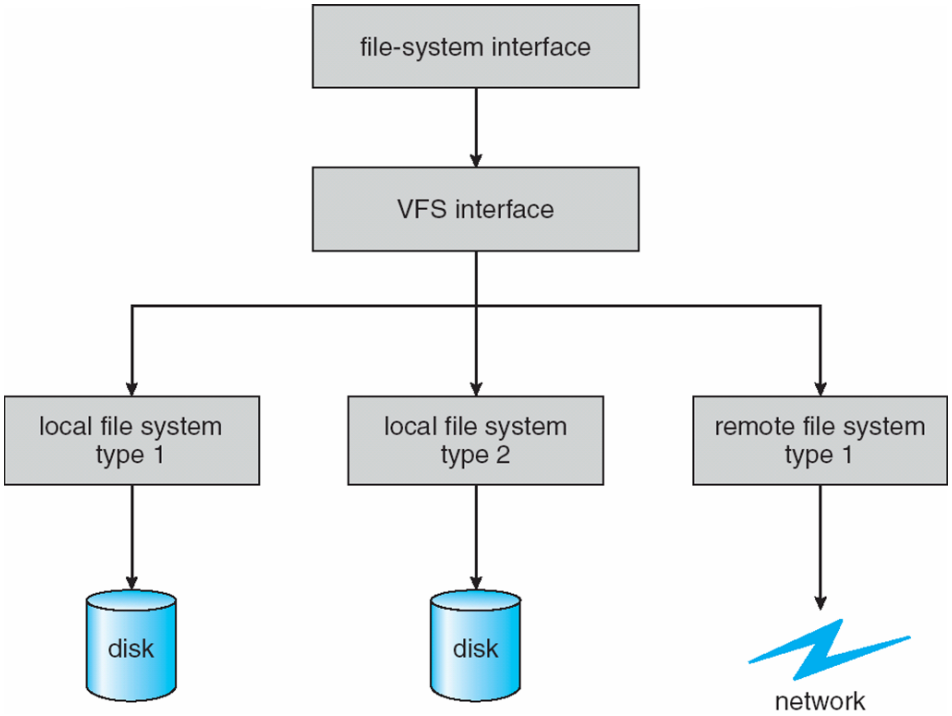
Virtual File Systems (**VFS**) on Unix provide an object-oriented way of implementing file systems

VFS allows the same system call interface (the API) to be used for different types of file systems

- Separates file-system generic operations from implementation details
- Implementation can be one of many file systems types, or network file system
  - Implements **vnodes** which hold inodes or network file details
- Then dispatches operation to appropriate file system implementation routines

Virtual File Systems (Cont.)

The API is to the VFS interface, rather than any specific type of file system



Virtual File System Implementation

For example, Linux has four object types:

inode, file, superblock, dentry

VFS defines set of operations on the objects that must be implemented

Every object has a pointer to a function table

- Function table has addresses of routines to implement that function on that object
- For example:
- • `int open(. . .)`—Open a file
- • `int close(. . .)`—Close an already-open file
- • `ssize t read(. . .)`—Read from a file
- • `ssize t write(. . .)`—Write to a file
- • `int mmap(. . .)`—Memory-map a file

The Sun Network File System (NFS)

An implementation and a specification of a software system for accessing remote files across LANs (or WANs)

The implementation is part of the Solaris and SunOS operating systems running on Sun workstations using an unreliable datagram protocol (UDP/IP protocol and Ethernet

NFS (Cont.)

Interconnected workstations viewed as a set of independent machines with independent file systems, which allows sharing among these file systems in a transparent manner

A remote directory is mounted over a local file system directory

- The mounted directory looks like an integral subtree of the local file system, replacing the subtree descending from the local directory

Specification of the remote directory for the mount operation is nontransparent; the host name of the remote directory has to be provided

- Files in the remote directory can then be accessed in a transparent manner

Subject to access-rights accreditation, potentially any file system (or directory within a file system), can be mounted remotely on top of any local directory

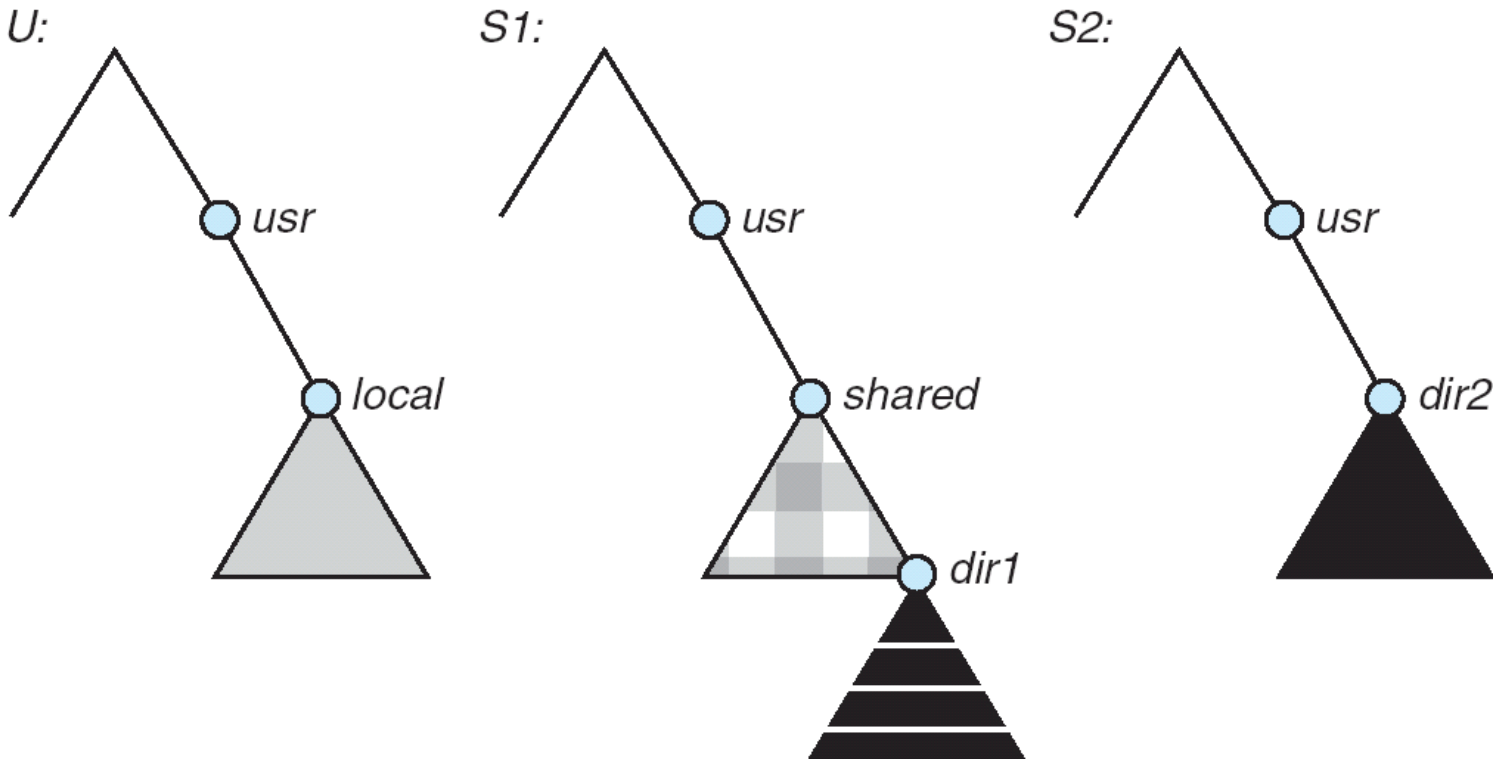
NFS (Cont.)

NFS is designed to operate in a heterogeneous environment of different machines, operating systems, and network architectures; the NFS specifications independent of these media

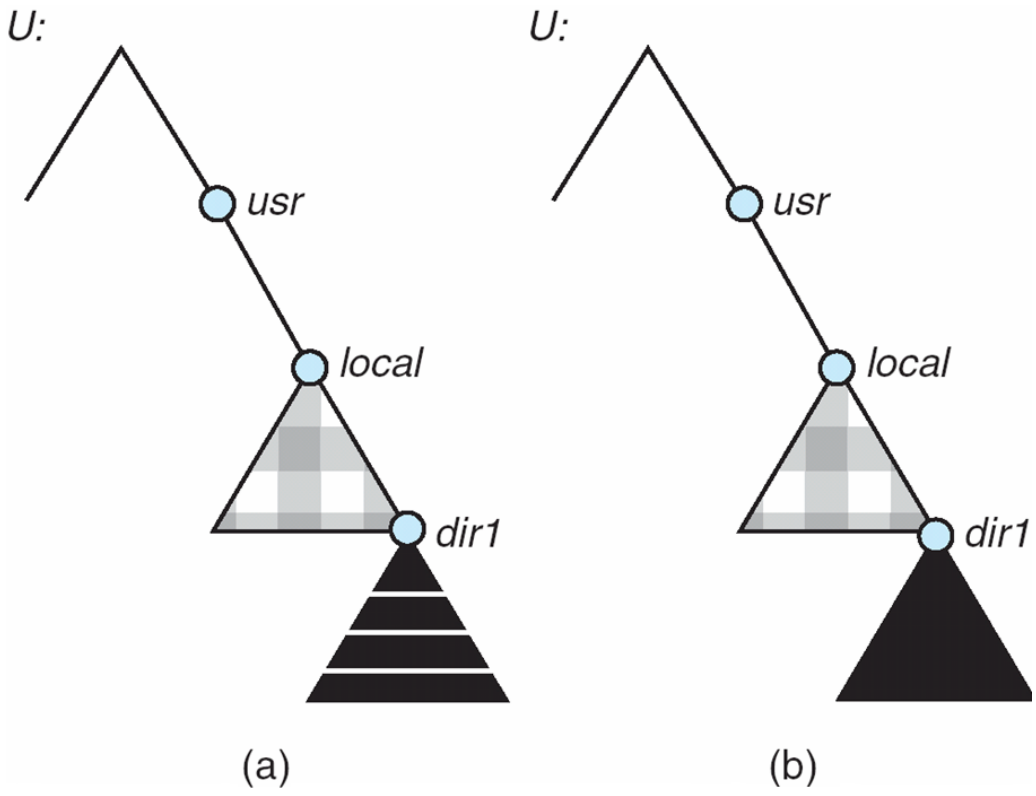
This independence is achieved through the use of RPC primitives built on top of an External Data Representation (XDR) protocol used between two implementation-independent interfaces

The NFS specification distinguishes between the services provided by a mount mechanism and the actual remote-file-access services

Three Independent File Systems



Mounting in NFS



Mounts

Cascading mounts

NFS Mount Protocol

- Establishes initial logical connection between server and client
- Mount operation includes name of remote directory to be mounted and name of server machine storing it
  - Mount request is mapped to corresponding RPC and forwarded to mount server running on server machine
  - Export list – specifies local file systems that server exports for mounting, along with names of machines that are permitted to mount them
- Following a mount request that conforms to its export list, the server returns a file handle—a key for further accesses
- File handle – a file-system identifier, and an inode number to identify the mounted directory within the exported file system
- The mount operation changes only the user's view and does not affect the server side

NFS Protocol

Provides a set of remote procedure calls for remote file operations. The procedures support the following operations:

- searching for a file within a directory
- reading a set of directory entries
- manipulating links and directories
- accessing file attributes
- reading and writing files

NFS servers are **stateless**; each request has to provide a full set of arguments (NFS V4 is just coming available – very different, stateful)

Modified data must be committed to the server's disk before results are returned to the client (lose advantages of caching)

The NFS protocol does not provide concurrency-control mechanisms

Three Major Layers of NFS Architecture

UNIX file-system interface (based on the **open, read, write**, and **close** calls, and **file descriptors**)

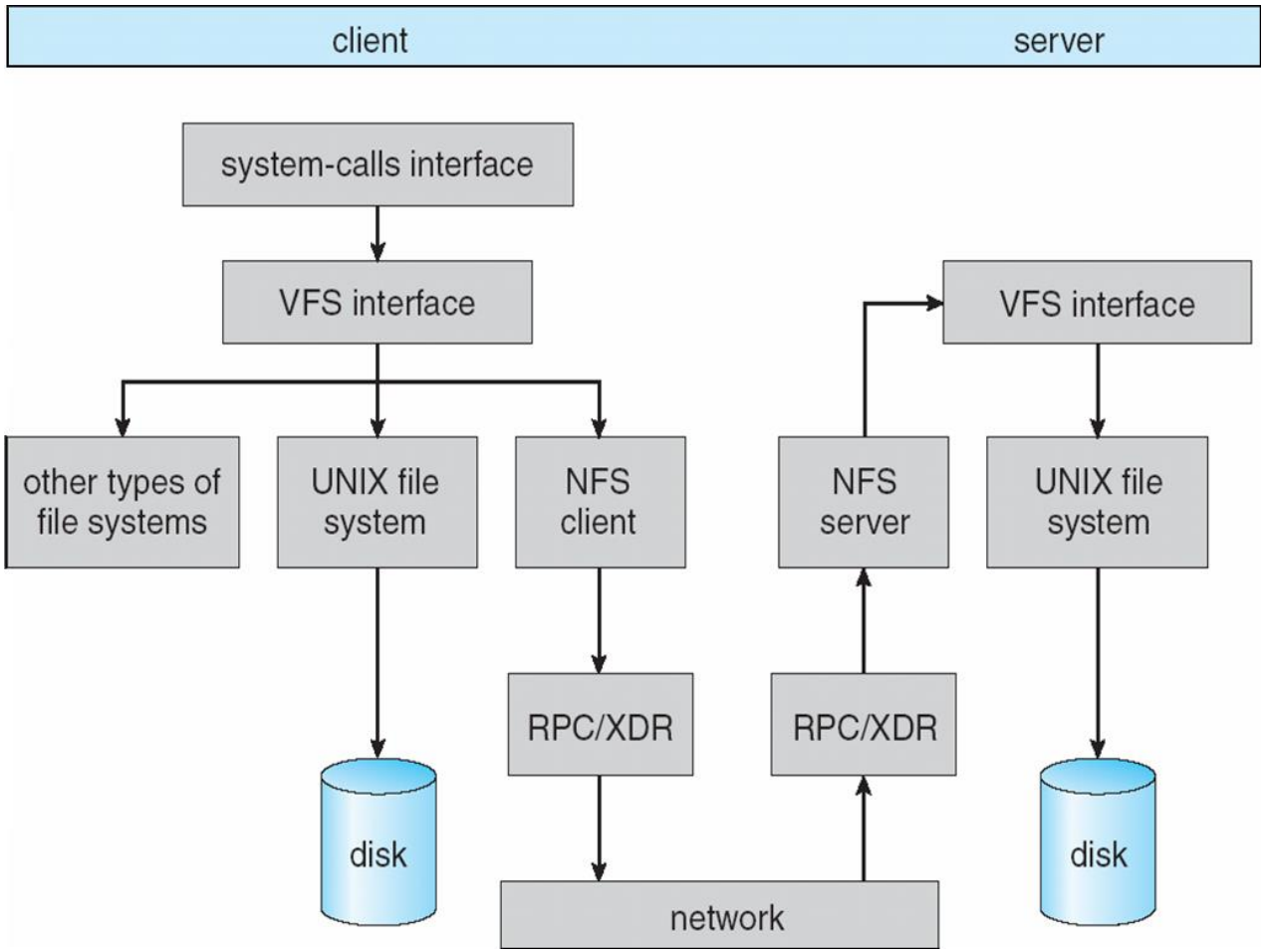
Virtual File System (VFS) layer – distinguishes local files from remote ones, and local files are further distinguished according to their file-system types

- The VFS activates file-system-specific operations to handle local requests according to their file-system types
- Calls the NFS protocol procedures for remote requests

NFS service layer – bottom layer of the architecture

- Implements the NFS protocol

Schematic View of NFS Architecture



NFS Path-Name Translation

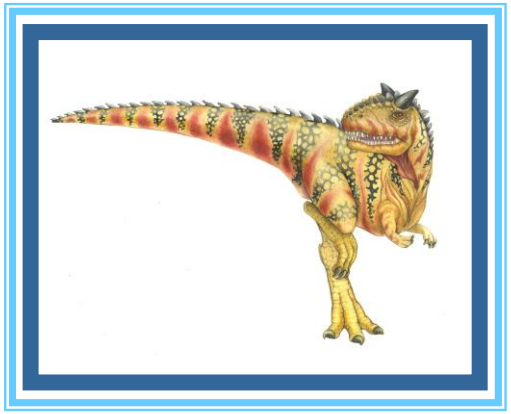
Performed by breaking the path into component names and performing a separate NFS lookup call for every pair of component name and directory vnode

To make lookup faster, a directory name lookup cache on the client's side holds the vnodes for remote directory names

NFS Remote Operations

- Nearly one-to-one correspondence between regular UNIX system calls and the NFS protocol RPCs (except opening and closing files)
- NFS adheres to the remote-service paradigm, but employs buffering and caching techniques for the sake of performance
- File-blocks cache – when a file is opened, the kernel checks with the remote server whether to fetch or revalidate the cached attributes
  - Cached file blocks are used only if the corresponding cached attributes are up to date
- File-attribute cache – the attribute cache is updated whenever new attributes arrive from the server
- Clients do not free delayed-write blocks until the server confirms that the data have been written to disk

End of Chapter 15



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